

Lecture 28

Frequentist properties of Bayesian methods

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(Edited from the lecture notes by Brian J. Reich, NC State)



Calibrated Bayes

- ▶ So far we have discussed Bayesian methods as being separate from the frequentist approach.
- ▶ However, in many cases, methods with frequentist properties are desirable.
- ▶ For example, we may want a method with Type-I error control or 80% power.
- ▶ You can design Bayesian methods to achieve these frequentist properties.
- ▶ In this view, Bayesian methods generate procedures/algorithms for further study.
- ▶ Often Bayesian methods are very competitive with frequentist methods using frequentist criteria.

Outline of Chapter 7

- ▶ Decision theory
- ▶ Bias-variance tradeoff
- ▶ Asymptotics
- ▶ Simulation studies

Should Bayesians care about frequentist properties?

What if a Bayesian weather forecaster made a 95% prediction interval for temperature every day for a year but the interval only included the actual temperature 40% of the time?

Calibrated Bayes, for Statistics in General, and Missing Data in Particular¹

Roderick Little

Abstract. It is argued that the Calibrated Bayesian (CB) approach to statistical inference capitalizes on the strength of Bayesian and frequentist approaches to statistical inference. In the CB approach, inferences under a particular model are Bayesian, but frequentist methods are useful for model development and model checking. In this article the CB approach is outlined. Bayesian methods for missing data are then reviewed from a CB perspective. The basic theory of the Bayesian approach, and the closely related technique of multiple imputation, is described. Then applications of the Bayesian approach to normal models are described, both for monotone and nonmonotone missing data patterns. Sequential Regression Multivariate Imputation and Penalized Spline of Propensity Models are presented as two useful approaches for relaxing distributional assumptions.

Key words and phrases: Maximum likelihood, multiple imputation, penalized splines, propensity models, sequential regression multivariate imputation.

Little in *Little, 2011, Stat Sci*

- ▶ Bayesian statistics is strong for inference under an assumed model, but relatively weak for the development and assessment of models.
- ▶ Frequentist statistics provides useful tools for model development and assessment, but has weaknesses for inference under an assumed model.
- ▶ If this summary is accepted, then the natural compromise is to use frequentist methods for model development and assessment, and Bayesian methods for inference under a model.
- ▶ This capitalizes on the strengths of both paradigms, and is the essence of the approach known as Calibrated Bayes.

Rubin on *Little, 2011, Stat Sci*

- ▶ The applied statistician should be Bayesian in principle and calibrated to the real world in practice - appropriate frequency calculations help to define such a tie.
- ▶ Frequency calculations are useful for making Bayesian statements scientific, scientific in the sense of capable of being shown wrong by empirical test.
- ▶ Here the technique is the calibration of Bayesian probabilities to the frequencies of actual events.

Bayes as a procedure generator

- ▶ A Bayesian analysis produces a posterior distribution which summarize our uncertainty after observing the data.
- ▶ However, if you have to give a one-number summary as an estimate you might pick the posterior mean

$$\hat{\theta}_B = E(\theta|\mathbf{Y}).$$

- ▶ This estimator $\hat{\theta}_B$ can be evaluated along with MLE or method of moments estimators.
- ▶ Is it biased? Consistent? How does its MSE compare with the MLE?
- ▶ These are all frequentist properties of the Bayesian estimator.

Bayes as a procedure generator

- ▶ Similarly, if we have to give an interval estimate, we might use the 95% posterior credible set.
- ▶ In practice, this interval is motivated by the one data set we observed.
- ▶ But we could view this as a procedure for constructing an interval and inspect its frequentist properties.
- ▶ If we analyzed many datasets, each time computing a 95% posterior interval, how many would contain the true value?
- ▶ A Bayes test is to reject H_0 if $\text{Prob}(H_0|\mathbf{Y}) < c$.
- ▶ What are the Type I and Type II errors of this test?
- ▶ Can we pick the threshold c to control Type I error?

Bayesian decision theory

- ▶ Before studying the frequentist properties of Bayesian estimators and hypothesis tests, we should determine the “best” Bayesian method.
- ▶ For example, should we take the estimator to be the posterior mean, median, or mode?
- ▶ Defining “best” requires a scoring system.
- ▶ We call this the loss function $l(\hat{\theta}, \theta)$.
- ▶ Squared error loss is $l(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$.
- ▶ Absolute loss is $l(\hat{\theta}, \theta) = |\hat{\theta} - \theta|$.

Bayesian decision theory

- ▶ The summary of the posterior that minimizes the expected (posterior) loss is the **Bayes rule**.
- ▶ Squared error loss implies we should use the posterior mean for $\hat{\theta}$.
- ▶ Absolute loss implies we should use the posterior median for $\hat{\theta}$.
- ▶ Hypothesis test requires are more complicated loss function.

Squared error loss

Let $\bar{\theta} = E_{\theta|Y}(\theta)$ denotes the posterior mean.

$$\begin{aligned}E_{\theta|Y}[I(\hat{\theta}, \theta)] &= E_{\theta|Y}[(\hat{\theta} - \theta)^2] \\&= E_{\theta|Y}[(\hat{\theta} - \bar{\theta} + \bar{\theta} - \theta)^2] \\&= E_{\theta|Y}[(\hat{\theta} - \bar{\theta})^2] + 2E_{\theta|Y}[(\hat{\theta} - \bar{\theta})(\bar{\theta} - \theta)] + E_{\theta|Y}[(\bar{\theta} - \theta)^2] \\&= (\hat{\theta} - \bar{\theta})^2 + 2(\hat{\theta} - \bar{\theta})E_{\theta|Y}[(\bar{\theta} - \theta)] + E_{\theta|Y}[(\bar{\theta} - \theta)^2] \\&= (\hat{\theta} - \bar{\theta})^2 + V_{\theta|Y}[\theta]\end{aligned}$$

Here $V_{\theta|Y}[\theta]$ does not depend on θ . Thus, $E_{\theta|Y}[I(\hat{\theta}, \theta)]$ is minimized when $\hat{\theta} = \bar{\theta}$.

Absolute error loss

We want to minimize $E_{\theta|Y}(|\hat{\theta} - \theta|) = \int |\hat{\theta} - \theta| \pi(\theta|Y) d\theta$.

$$\begin{aligned} E_{\theta|Y}(|\hat{\theta} - \theta|) &= \int |\hat{\theta} - \theta| \pi(\theta|Y) d\theta \\ &= \int_{-\infty}^{\hat{\theta}} |\hat{\theta} - \theta| \pi(\theta|Y) d\theta + \int_{\hat{\theta}}^{\infty} |\hat{\theta} - \theta| \pi(\theta|Y) d\theta \\ &= \int_{-\infty}^{\hat{\theta}} (\hat{\theta} - \theta) \pi(\theta|Y) d\theta + \int_{\hat{\theta}}^{\infty} (\theta - \hat{\theta}) \pi(\theta|Y) d\theta \end{aligned}$$

Differentiate both the sides with respect to $\hat{\theta}$ and equate to zero. We get

$$\int_{-\infty}^{\hat{\theta}} \pi(\theta|Y) d\theta = \int_{\hat{\theta}}^{\infty} \pi(\theta|Y) d\theta$$

Thus, $\hat{\theta}$ is the median of $\pi(\theta|Y)$.

Bias/variance trade-off

- ▶ Assume $Y_1, \dots, Y_n \sim \text{Normal}(\mu, \sigma^2)$.
- ▶ Estimator 1: $\hat{\mu}_1 = \bar{Y}$.
- ▶ Estimator 2: $\hat{\mu}_2 = c\bar{Y}$ where $c = \frac{n}{n+m}$.
- ▶ $\hat{\mu}_2$ is the posterior mean under prior $\mu \sim \text{Normal}(0, \frac{\sigma^2}{m})$.
- ▶ Compute the bias and variance of each estimator.
- ▶ Compute the mean squared error (recall $\text{MSE} = \text{bias}^2 + \text{variance}$).
- ▶ Which estimator is preferred?

Properties of Bayesian estimators

Broadly speaking, the following comparisons between Bayes and MLE hold:

- ▶ Bayesian estimators have smaller standard errors because the prior adds information.
- ▶ Bayesian estimators are biased if the prior is not centered on the truth.
- ▶ Depending on this bias/variance trade-off, Bayes estimators may have smaller MSE than the MLE.
- ▶ If the prior is weak the methods are similar.
- ▶ For any prior that does not depend on the sample size, as n increases the prior is overwhelmed by the likelihood and the posterior approaches the MLE's sampling distribution.

Bayesian central limit theorem

- ▶ Assumptions:

- ▶ the usual MLE conditions on the likelihood.
- ▶ the prior does not depend on n and puts non-zero probability on the true value θ_0 .

- ▶ Then

$$p(\theta|\mathbf{Y}) \rightarrow \text{Normal} \left[\theta_0, I(\theta_0)^{-1} \right],$$

where I is the information matrix.

- ▶ Therefore, for large datasets the posterior is approximately normal.
- ▶ Bayes methods are asymptotically unbiased.

Bayesian central limit theorem

- ▶ This implies that Bayes and MLE will be equivalent in large samples.
- ▶ What a relief!
- ▶ However, the interpretation is different.
- ▶ We can use the Bayesian interpretation like $\text{Prob}(\mathcal{H}_0|\mathbf{Y})$ and $\text{Prob}(3.4 < \theta < 5.6)$.
- ▶ The Bayesian CLT gives a way to approximate ($n \rightarrow \infty$) the posterior without MCMC.
- ▶ Most still use MCMC with the hope that it better approximates ($S \rightarrow \infty$) the exact posterior.
- ▶ The CLT is useful for initial values and tuning.

Methods for studying frequentist properties

- ▶ Theoretical studies of Bayesian estimators use the same basic approaches as frequentist methods.
- ▶ Theorems and proofs (of consistency etc.) are ideal.
- ▶ When the math is intractable, simulation studies are used.
- ▶ In a simulation study you generate many datasets with known parameters values.
- ▶ You apply the Bayesian method to each dataset (so you may have to run MCMC several times).
- ▶ You then see how you did, e.g., what proportion of the 95% credible sets included the true value?

OLS versus Bayesian LASSO

n	p_0	p_1	MSE		Coverage	
			OLS	BLR	OLS	BLR
40	20	0	5.40	0.03	94.7	100.0
	15	5	5.71	3.45	93.8	96.0
	0	20	5.40	9.47	93.7	91.6
100	20	0	1.17	0.02	95.8	100.0
	15	5	1.27	0.98	94.5	95.5
	0	20	1.22	1.26	96.0	95.6

- ▶ n is the sample size.
- ▶ p_0 is the number of null covariates with $\beta_j = 0$.
- ▶ p_1 is the number of non-null covariates with $\beta_j \neq 0$.

Methods for studying frequentist properties

Conclusions:

- ▶ When the model is sparse (p_1 is small), BLR is has much smaller MSE than OLS.
- ▶ When the model is dense (p_0 is small), OLS has smaller MSE, but for large n the methods are similar.
- ▶ Both methods generally have reasonable coverage.
- ▶ BLR's coverage is low when n is small and the model is dense, i.e., when its assumptions are grossly violated.

Thank you!